same face pointing toward Gargantua (Figure 17.4), or nearly so (as I discuss later).

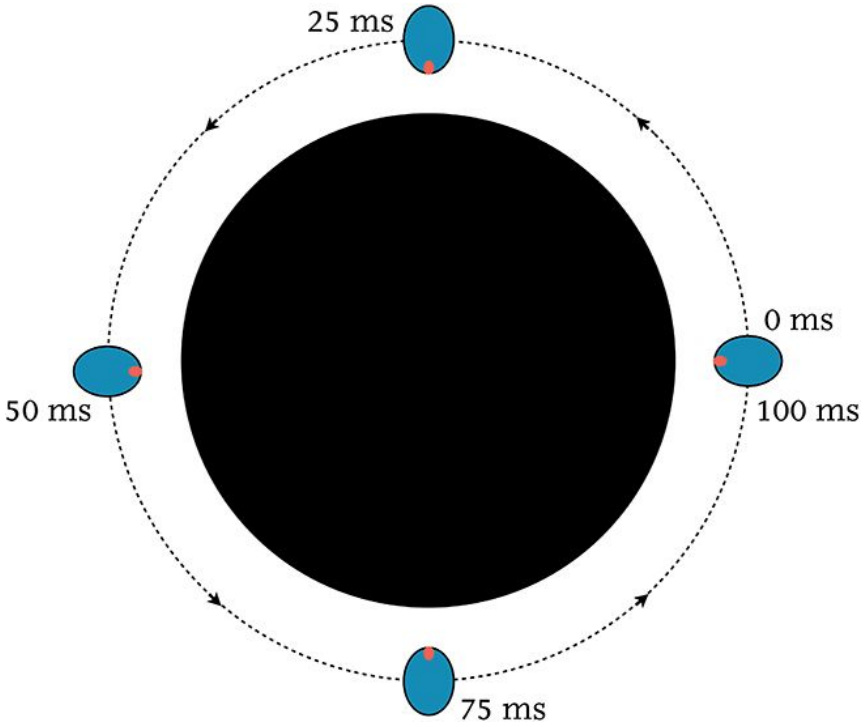


Fig. 17.4. The orbital motion and spin of Miller's planet relative to distant stars. The red spot on the planet's surface and the tidal bulge always face Gargantua.

The Whirl of Space

Einstein's laws dictate that, as seen from afar, for example, from Mann's planet, Miller's planet travels around Gargantua's billion-kilometer-circumference orbit once each 1.7 hours. This is roughly half the speed of light! Because of time's slowing, the Ranger's crew measure an orbital period sixty thousand times smaller than this: a tenth of a second. Ten trips around Gargantua per second. That's *really* fast! Isn't it far faster than light? No, because of the space whirl induced by Gargantua's fast spin. Relative to the whirling space at the planet's location, and using time as measured there, the planet is moving slower than light,